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Developing a local food business model and innovation framework: a systematic literature review

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Local food production has been identified as a key lever for advancing sustainable and resilient food systems, yet its development remains limited in scope. To address this challenge, scholars have increasingly examined local food businesses through a business model (BM) perspective, which provides insights into producers' value intentions and how they propose, create, deliver, and capture value. However, research on these BM components and their potential for innovation has thus far been fragmented and not comprehensively examined. In response, this study conducts a systematic literature review, selecting 44 articles in accordance with PRISMA guidelines. Based on this review, the article proposes a local food BM and innovation framework, that includes (i) a comprehensive overview of BM components in the local food context, (ii) preconditions, approaches and 10 strategies for innovating local food BMs, and (iii) a discussion of the interdependencies and multidimensional relationships between BM components and innovation strategies. Finally, the article outlines future research directions and managerial implications to guide both scholars and practitioners in strengthening local food production and advancing the transition towards sustainable food systems.

KEYWORDS

local food, business model, innovation, food systems, systematic literature review

1 Introduction

Governments and civil society organisations are increasingly calling for market diversification and the restructuring of food systems to address challenges associated with global food systems, such as resource depletion and food insecurity (European Commission (EC), 2024; Jones et al., 2022; McGreevy et al., 2022). One widely discussed pathway towards more sustainable and resilient systems is strengthening local food production, which has been emphasised for its potential to improve the economic viability of producers, enhance social equity and enable more environmentally friendly practices (Donner and de Vries, 2023; Enthoven and den Broeck, 2021; Michel-Villarreal et al., 2019).

Scholars have approached the definition and establishment of local food production in various ways, often drawing on the concept of alternative food networks (AFN) and small-scale producers (Michel-Villarreal et al., 2019; Savarese et al., 2020). Central to these perspectives are different forms of proximity, geographical and social proximity between producers and customers, as well as proximity in values such sustainability or quality (Eriksen, 2013; Nilsson, 2009; Pardillo Baez et al., 2020). In addition, scholars have examined key actors and resource flows through local, short or value-based supply chains (De Rosa et al., 2024; Saleh et al., 2025), as well as opportunities to scale local food businesses and access new markets (Fleiß and Aggestam, 2016; Mars, 2022). Despite this, local food production remains limited in scope with markets still dominated by large food companies, leaving local food producers struggling to compete (Esbjerg et al., 2014; Hersleth et al., 2023; Jones et al., 2022). Consequently, scholars emphasise the need to better understand how local food business generate value, namely through

their business model (BM) (Michel-Villarreal et al., 2019; Sadovska et al., 2023), and to place more focus on entrepreneurial strategies and BM innovation (Barth et al., 2017; Hersleth et al., 2023; Tell et al., 2019).

The BM perspective is increasingly applied in agri-food research to advance discussions on sustainable food systems (Barth et al., 2017; Donner and de Vries, 2023; Kumar et al., 2022). A BM describes the logic of how an organisation conducts business, which can be a starting point for helping organisations to analyse and compare business performance, management and innovation (Osterwalder et al., 2005). Sustainable BM literature often adopts a value perspective, consisting of four components: (1) value intention, referring to the entrepreneur's attitude and motivation; (2) value proposition, describing what value is offered and to whom (Barth et al., 2017; Bocken et al., 2015); (3) value creation and delivery, referring to the key activities, resources, distribution channels, key partners and relationships, and technologies used, matching the value proposition; and (4) value capture, encompassing the business' cost structure and revenue streams, as well as value captured by key actors including how natural and social values are sustained and regenerated (Bocken et al., 2015; Richardson, 2008). Scholars further stress the importance of engaging a wide range of stakeholders across all BM components and in BM innovation (Fobbe and Hilletofth, 2021). BM innovation is closely linked with the value intention and can involve transforming one or several BM components, but it may also include diversifying the BM by adding a new BM to the existing BM, starting up an entirely new BM, or acquiring a company with a different BM (Barth et al., 2017; Clauss, 2017; Geissdoerfer et al., 2018).

Recent literature reviews highlight local food BMs as important parts of sustainable food systems and explore how local BM features are associated with sustainable food BMs (Donner and de Vries, 2023; Kumar et al., 2022; Miranda et al., 2023); however, they consider local food as a sub-category among others such as circular BMs focused on reducing waste, or sufficiency BMs encouraging sustainable consumption. Further, research on local food BM and BM innovation is often focused on a specific type of BM (Belliggiano et al., 2024; Wiśniewska-Paluszak et al., 2023) or specific BM components (Bauman et al., 2019; Sebök et al., 2022), segmenting the literature. This makes it difficult to understand the overarching logic of how local food businesses operate and to compare different local food BMs (Michel-Villarreal et al., 2019). Indeed, scholars critique that the BM perspective is still not sufficiently developed in the literature where the value components of local food BMs have not yet been comprehensively examined (Michel-Villarreal et al., 2019; Miranda et al., 2023).

Against this background, the aim of this paper is to advance the debate on local food by systematically reviewing literature on local food BMs and BM innovation, structuring and synthesising findings into a comprehensive framework, and identifying knowledge gaps and avenues for future research. Thus, rather than compiling a list of BM types, the study seeks to uncover the overarching logic of how local food businesses operate and how their BMs can be innovated, based on a systematic literature review. Specifically, it addresses two research questions: (RQ1) What do the value components of local food BMs encompass? (RQ2) How can local food BMs be innovated?

2 Methods

Systematic literature reviews (SLRs) are a common research method to assess, evaluate and synthesise the existing body of knowledge in a robust and transparent manner, allowing to recognise patterns and themes and uncover future research direction (Linnenluecke et al., 2020; Tranfield et al., 2003). To conduct this review, the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) approach was followed (Page et al., 2021). This approach ensures systematic screening, assessment and selection of relevant and high-quality literature and has been widely applied in SLRs within agri-food literature (for example, Donner and de Vries, 2023).

2.1 Search strategy

To search for and select relevant literature, a comprehensive search string based on two keyword themes, namely "business model" and "local food", was used (Linnenluecke et al., 2020). These keywords, along with synonyms and relevant variations were included in the search strings (Table 1). The search strings were linked with the Boolean operator "AND."

The search was conducted in two databases, SCOPUS and Web of Science, until the 11th of February 2025. Using two databases may create several duplicates in the search results, but it also enhances the rigour of the search as no single database contains all published material (Linnenluecke et al., 2020; Xiao and Watson, 2019). The databases were chosen as they have been recognised for being one of the largest databases of peer-reviewed literature as well as for including high-quality journals and providing advanced search functions (Gusenbauer and Haddaway, 2020). In addition, these databases have been widely used in SLRs on related topics (Donner and de Vries, 2023; Michel-Villarreal et al., 2019; Miranda et al., 2023). The search in the two databases yielded 1,235 articles, of which 722 were identified in Scopus and 513 in Web of Science (Figure 1).

2.2 Screening and data extraction

In the screening stage, articles were first included or excluded based on the article type and language (Table 2), leading to the exclusion of 399 articles. Then all remaining 836 records were exported to the software ryyan to detect duplicates and support the assessment and selection process. In total, 278 articles were duplicates and subsequently removed, leading to 558 articles. From those articles, the title, abstract and keywords were screened, applying inclusion criteria derived from the research questions. Articles dealing with local food BM related to other research areas, such as consumption,

TABLE 1 Search strings.

Keyword theme	Search strings
Business model	("business model*" OR "value creat*" OR "creat* value" OR "value capt*" OR "capt* value" OR "value propos*" OR "propos* value" OR "value deliver*" OR "deliver* value" OR "co-creat* value" OR "value co-creat*" OR "value destroy*" OR "destroy* value" OR "value intention*" OR "business model innovation")
Local food	("food*" AND ("local" OR "short" OR "alternative" OR "value based"))

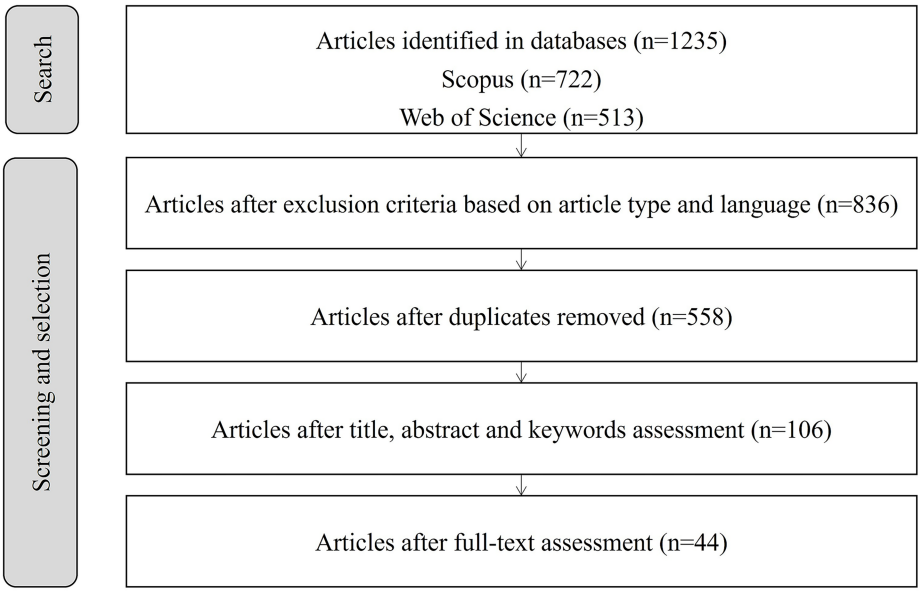


FIGURE 1
Results of the systematic literature review based on PRISMA protocol (Page et al., 2021).

TABLE 2 Inclusion and exclusion criteria.

Criterion	Inclusion	Exclusion
Article type	Peer-reviewed journal article Empirical, conceptual	Conference papers, books, book chapters, editorials, literature reviews, proceedings
Language	English	Any other language
Focus	Focused on BMs, BM components, or BM innovation in local food context	Not relevant for research questions Does not include a producer perspective, technical focus

retail, waste, or tourism were excluded if they did not contain a producer perspective. Further, articles with a pure technical focus such as vertical farming were excluded. 452 articles were evaluated as false positives in the abstract screening as they referred to the topics of BM or local food, but only tangentially dealt with the BM components or BM innovation focused on local food (Linnenluecke et al., 2020). In the final step, the remaining 106 articles were read in full and a final decision to include or exclude was made based on the relevance to answer the research questions (Linnenluecke et al., 2020; Tranfield et al., 2003). Forty-four articles were selected for inclusion in this study after full-text assessment. An overview of the articles included can be found in Supplementary material.

In the data extraction stage, all relevant information from the included studies was extracted into Microsoft Excel for a descriptive analysis to provide a comprehensive overview of the field of study. Thus, the main formal characteristics of each article were analysed, including year of publication, journal and research methods applied (Tranfield et al., 2003). This allowed to determine research trends and diversity in the field. Next, a manual qualitative content analysis of the included articles was conducted to synthesise the articles and code their content. Data was first analysed deductively using a coding scheme that was based on BM literature. Following this, an inductive analysis of the included article was conducted focused on the individual BM components and their specific elements, as well as on BM innovation, where categories such as preconditions and approaches for BM innovation as well as strategies to innovate the BM evolved (Figure 2). During this process, it was noted that articles use different terms, such as producer, entrepreneur and farmer, as well as customer and consumer. To improve the clarity of this paper, only the term producer is used to refer to the owner of the local food business, and the term customer to refer to the person buying the product, independent if they consume it or further sell it. Further, it has to be noted that many articles contained information for multiple BM components or BM innovation; therefore, articles are referred to multiple times.

2.3 Overview of identified studies

The selected articles were published between 2008 and 2024; however, the number of articles only started to increase significantly after 2021, indicating a growing interest in the topic. The majority of the articles reviewed adopted a case study approach (23), followed by multiple methods (9), and survey research (6), implying that the BM perspective within local food context is still an emerging research field. Less studies used conceptual research (2), action research (2), or other approaches (2).

The articles included in this review have been published in 34 different journals across a variety of research domains, with most articles being published in journals with a focus on food and agriculture (35.3%) indicating a growing interest in the topic of BMs by food scholars while there were less articles published in journals focused on business and management (23.5%), sustainability or land and rural studies (each 11.8%), or journals with another focus (17.6%). Most articles included in this study were published in *British Food Journal* (4), followed by *Sustainability* and *Agriculture* (each 3 articles).

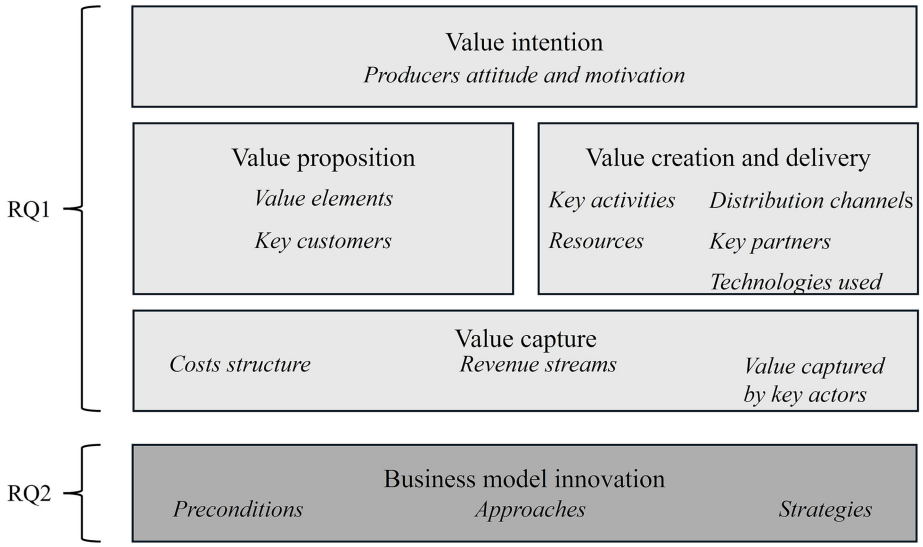


FIGURE 2
Data extraction categories.

3 Results

3.1 Local food BM components

3.1.1 Value intention

Local food producers tend to be mainly motivated by economic viability (Albrecht and Smithers, 2018; Recasens et al., 2016; Wiśniewska-Paluszak and Paluszak, 2021; Wubben et al., 2013). This may be related to the struggles for economic profitability and feeling of lack of control over the business when participating in conventional food supply chains (Albrecht and Smithers, 2018). However, scholars highlight that the value intention is often multidimensional, and producers can also be motivated by environmental and social concerns as well as the satisfaction and appreciation derived from producing local food (Albrecht and Smithers, 2018; Brettin, 2024; Schlett et al., 2024). This can encompass the motivation to reduce food waste or food miles, or to disseminate local knowledge and culture, facilitate connections in the local community and promote the consumption of traditional products (De Rosa et al., 2024; Savarese et al., 2020; Wiśniewska-Paluszak and Paluszak, 2021). Brettin (2024) describes this as a network of responsibilities, where the responsibility to ensure the viability of the business, and responsibilities to produce high-quality food, avoid food waste, generate socio-cultural values, and ensure animal welfare and environmental sustainability are intertwined. The intensity of the different motivations can however differ, for example a study by Recasens et al. (2016) highlighted the connection of producers and customers, traditional land use, and leisure as important motivations whereas environmental sustainability, transparency or education were found to be less important. Further, producers that are deeply connected to their local environment and community were found to be more likely to prioritise sustainability issues as part of their production processes (Schlett et al., 2024).

3.1.2 Value proposition

3.1.2.1 Value proposition elements

The review showed that there is a large diversity of elements that can encompass a local food value proposition, which can be categorised into proximity and place-making values, product characteristics values, intangible values and sustainability values.

Proximity values, often related to close relationships and limited geographical distance between producers and customers, have been recognised as important part of the value proposition (Drejerska et al., 2019; Moschitz and Oehen, 2020). This can allow customers to not only buy a product in close geographic proximity but to familiarise themselves with the people behind it (De Bernardi and Tirabeni, 2018; Hersleth et al., 2023; Mars, 2022; Thomas-Francois et al., 2018; Wiśniewska-Paluszak and Paluszak, 2021). In addition, scholars argue that local food value propositions by proximity are often intertwined with local food value propositions by place-making. This can encompass having a clear territorial reference in the value proposition or embedding specific localised values in a product linked to supporting and maintaining the local agrobiodiversity (Di Gregorio, 2017; Moschitz and Oehen, 2020; Radić et al., 2023). A place-making value proposition is thus not limited by geographical proximity, allowing to connect with wider customer groups from distant territories attracted by the value embedded in the product (Di Gregorio, 2017; Moschitz and Oehen, 2020; Radić et al., 2023).

Several scholars also highlight specific product characteristics as core of the value proposition (Charatsari et al., 2023; Drejerska et al., 2019; Sebök et al., 2022). This includes high-quality, local handicraft, natural ingredients, freshness, organic certifications as well as authentic taste (Drejerska et al., 2019; Hersleth et al., 2023; Prosperi et al., 2023; Sebök et al., 2022). The quantity and availability of local food products have also been discussed as part of the value proposition. For example, scholars argue that offering limited and seasonal products belonging to the territory in favour of quality and freshness is

important to differentiate the product from industrial products and to enhance the value (Drejerska et al., 2019; Radić et al., 2023).

In addition, a local food value proposition can offer intangible values linked to the production and consumption of the food (De Bernardi and Tirabeni, 2018; De Rosa et al., 2024). These can be influenced by the culture and rural traditions and can include local skills, knowledge and experiences shaping the product or service offerings (De Bernardi and Tirabeni, 2018; Hersleth et al., 2023; Thomas-Francois et al., 2018; Wiśniewska-Paluszak et al., 2023). Further, authenticity and integrity, trust, and maintenance of values such local cultural heritage are important to connect with customers (Albrecht and Smithers, 2018; Radić et al., 2023).

Finally, scholars argue that local food BMs need to be sustainable with respect to environmental and social wellbeing as well as economic resilience (De Bernardi and Tirabeni, 2018; Borsotto et al., 2023; Charatsari et al., 2023; Sebök et al., 2022). However, how specific sustainability values are embedded in the value proposition can vary depending on the value intention (De Bernardi and Tirabeni, 2018; Wiśniewska-Paluszak et al., 2023). For example, a value proposition focused on environmental values can include organic food, origin certifications, focus on products belonging to the territory and their seasonality, or production with reduced emissions (De Bernardi and Tirabeni, 2018; Hersleth et al., 2023; Prosperi et al., 2023; Sebök et al., 2022). In comparison, a value proposition oriented towards social values can emphasise working rights, animal welfare, strengthen local communities or promote healthy consumption and nutritional products (De Bernardi and Tirabeni, 2018; Charatsari et al., 2023; Prosperi et al., 2023; Sebök et al., 2022; Wiśniewska-Paluszak et al., 2023). Finally, the economic dimension can focus on fair price for producers or support for local communities (Sebök et al., 2022).

3.1.2.2 Key customers

Depending on the value proposition, local food businesses can attract a variety of customers. Customers are often clustered into distinct segments based on geographical, demographic, behavioural, and psychographic criteria (Drejerska et al., 2019), based on different demands and interest, or different consumption practices (Belliggiano et al., 2024; Pölling et al., 2017). Scholars typically discuss final customers and local buyers, including households from local communities, as well as restaurants and hotels, specialty stores, cooperatives, and retailers and wholesalers (Belliggiano et al., 2024; Charatsari et al., 2023; Cirone et al., 2023; Wiśniewska-Paluszak et al., 2023). In addition, governmental institutions such as schools or elderly houses are recognised as potential customers for local food BMs (Charatsari et al., 2023).

3.1.3 Value creation and delivery

3.1.3.1 Key activities

Scholars describe several key activities that are part of the value creation of local food BMs. These can be summarised into two core activities, encompassing farming, and the production and processing of products (Belliggiano et al., 2024; Charatsari et al., 2023; Drejerska et al., 2019; Prosperi et al., 2023). Further, four facilitating and supporting activities are discussed, these are marketing, selling of products including packaging and transport, payment transfers and demand planning (Belliggiano et al., 2024; Charatsari et al., 2023; Drejerska et al., 2019; Prosperi et al., 2023), and in some cases, purchase of raw materials and storage (Charatsari et al., 2023; Drejerska et al., 2019). While marketing is increasingly regarded as an important activity, Albrecht and Smithers (2018) also highlight challenges that stem from these new activities requiring new skills and taking up time. In addition, some scholars discuss other non-agricultural activities such as tourism, educational and cultural services or bio-energy production, highlighting the multifunctionality of local food BMs (Belliggiano et al., 2024; Rytkönen and Oghazi, 2022; Wiśniewska-Paluszak et al., 2023; Wiśniewska-Paluszak and Paluszak, 2021).

3.1.3.2 Resources

Various types of resources relevant for value creation in local food BMs are discussed by scholars. First, land, buildings and infrastructure are highlighted as important resources, in particular access to farmland and city proximity giving access to a large customer base (Charatsari et al., 2023; Pölling et al., 2017; Prosperi et al., 2023; Wiśniewska-Paluszak and Paluszak, 2021). At the same time, access to land and infrastructure is recognised as a challenge, mainly in terms of scaling, distribution, agricultural policies and urban development (Albrecht and Smithers, 2018; Keech et al., 2023; Sebök et al., 2022). Other resources related to land and infrastructure are irrigation resources and availability of local resources for production (Charatsari et al., 2023; Mars, 2015; Prosperi et al., 2023). For example, Mars (2015) discuss how supply scarcities both in terms of variety and quantity limit producers' capacity of producing products based on locally grown ingredients. This also influences price tags, highlighting a trade-off between locally sourced products and customers' willingness to pay for those products (Mars, 2015).

Further, working capital is discussed by scholars as local food BMs are often heavily based on voluntary and unpaid family-labour to support the often individually operating producer (Prosperi et al., 2023; Wiśniewska-Paluszak et al., 2023; Wiśniewska-Paluszak and Paluszak, 2021). This stems partly from the lack of availability of labour that producers must overcome (Sebök et al., 2022). In addition, financial resources and credits from banks allowing to make investments (Prosperi et al., 2023; Wiśniewska-Paluszak and Paluszak, 2021), as well as goods capital such as machinery and equipment, and plantings, seeds or fertilisers are highlighted as resources for local food BMs (Charatsari et al., 2023; Pölling et al., 2017; Wiśniewska-Paluszak and Paluszak, 2021). Last, social capital has been argued to be a key resource to implement a unique value proposition and enhance value creation. Social capital can include education and competence, knowledge and experience of the producers as well as shared goals, partnerships and networks with a specific community (Charatsari et al., 2023; Prosperi et al., 2023; Wiśniewska-Paluszak et al., 2023; Wiśniewska-Paluszak and Paluszak, 2021).

3.1.3.3 Distribution channels

Studies highlight that the value delivery logic of local food BMs is often a combination of two to three distribution channels (Alvarez et al., 2021; Belliggiano et al., 2024; Keech et al., 2023; Lawson et al., 2008; Recasens et al., 2016). Some scholars argue that using different distribution systems can help to diverge risk and allows for better planning and control (Mars, 2022; Prosperi et al., 2023); however, other scholars discuss that this can lead to inadequate level of sales, or inefficient contact between producer and final customer (Keech et al., 2023). In addition, there can be competition internally between

different distribution channels as well as cross-channel conflicts with other producers competing for the same final customers (Keech et al., 2023; Lawson et al., 2008). The distribution channels can be categorised into producer-led, buyer-led and distributor-led value delivery logic, with differences regarding the leading actor and the level of interaction between producers and final customer (Drejerska et al., 2019; Keech et al., 2023; Wubben et al., 2013).

Producer-led distribution can be direct or indirect and is typically discussed as the most common distribution channel (Belliggiano et al., 2024; Keech et al., 2023; Lawson et al., 2008; Recasens et al., 2016; Wiśniewska-Paluszak et al., 2023; Wiśniewska-Paluszak and Paluszak, 2021). Direct distribution is often the first market entry for producers and encompasses all channels where the producers sell directly to final customers, that is at the farm via farm shops or on-farm sales, or outside the farm via for example farmers markets or specific pick-up locations (Charatsari et al., 2023; Drejerska et al., 2019; Hersleth et al., 2023; Mars, 2022). These sales channels allow producers to communicate working methods and justify higher product costs to customers, build close relationships, and target new customers, while avoiding any kind of intermediaries (Brettin, 2024; Drejerska et al., 2019; Lawson et al., 2008; De Rosa et al., 2024). However, direct distribution is often unplanned, and in particular passers-by or farm-gate sales make up only a small percentage of the producers' sales (Schutzbank and Riseman, 2013). Examples of indirect producer-led distribution channels are web shops, e-commerce, subscription food boxes or vending machines (Belliggiano et al., 2024; Charatsari et al., 2023; Rytkönen and Oghazi, 2022), but scholars question the effectiveness of online and box sales in terms of costs for transporting the products to the customers (De Rosa et al., 2024). The different producer-led distribution channels require different planning, coordination and storage and are often strongly interlinked, for example, producers can offer their customers to order via an online storefront but pick-up the products at the farm or other pick-up sites (Mars, 2015; Schutzbank and Riseman, 2013).

Buyer-led distribution channels are led by final customers or intermediaries, and can encompass consumer cooperatives, community supported agriculture (CSA) as well as selling products via local specialty stores or local hotels, restaurants, and catering (HoReCa) (Belliggiano et al., 2024; Charatsari et al., 2023; Hersleth et al., 2023; Recasens et al., 2016; Savarese et al., 2020). These types of sales channels are typically based on planned sales and up-front payments, minimising producers' risk and stress of selling at markets (Keech et al., 2023; Prosperi et al., 2023; Schutzbank and Riseman, 2013). In addition, buyer-led distribution is often focused on supporting producers and establishing long-term relationships instead of point-of sale transactions (Savarese et al., 2020; Wubben et al., 2013). However, selling products via channels such as HoReCa requires good organisation and coordination, and time and energy to build relationships with for example chefs (Oleksiuk and Rull Quesada, 2023; Schutzbank and Riseman, 2013). This may explain why restaurants are less used than other sales channels (Schutzbank and Riseman, 2013).

Distributor-led channels can include producer cooperatives, local food platforms or selling via wholesalers and retailers (Charatsari et al., 2023; Di Gregorio, 2017; Hersleth et al., 2023; Keech et al., 2023). These channels are not necessarily local from a geographical perspective but provide local business owners with opportunities to connect with geographically dispersed markets (Belliggiano et al., 2024; Di Gregorio, 2017; Hersleth et al., 2023). At the same time, using these kinds of sale channels can pose a variety of challenges, for example, wholesalers and retailers can have requirements for certain prices, larger volumes or formal certifications that could lead to high costs (Hersleth et al., 2023; Schutzbank and Riseman, 2013). Studies are inconsistent how much these channels are used, with some indicating low percentage of sales to supermarkets and platforms (Alvarez et al., 2021; Lawson et al., 2008; Schutzbank and Riseman, 2013; Wiśniewska-Paluszak and Paluszak, 2021), while Recasens et al. (2016) found producers selling up to 30–40 percent of products to national or international markets using specialised distributors to facilitate the process. However, selling to distant markets can affect the value proposition and value creation activities as local value elements need to be reinforced by sales partners or certifications which in turn could affect the autonomy of the producer and authenticity of the product (Di Gregorio, 2017; Moschitz and Oehen, 2020). Thus, producers need to choose between delegating tasks such as transportation or selling to an intermediary, potentially reducing profit and control on how products are displayed, versus owning their distribution channels which can be a time-consuming process that requires time away from the core activities (Albrecht and Smithers, 2018; Rytkönen and Oghazi, 2022).

3.1.3.4 Key partners and relationships

Scholars highlight how producers are embedded in their local environment, needing to engage and collaborate with a variety of stakeholders and building long-term and trust-based relationships for value creation and delivery (Pölling et al., 2017; Stoeva et al., 2024; Wiśniewska-Paluszak et al., 2023; Wubben et al., 2013). This review identified customers, suppliers, other producers, and other actors as key partners.

The relationship between producers and customers has been highlighted as key for being able to create and deliver value (Mars, 2015, 2022; Savarese et al., 2020). For example, scholars highlight the role customers play in helping producers understanding demand or in taking over roles such as marketers for the business, making them eventually co-creators of value (Mars, 2022; Radić et al., 2023; Stoeva et al., 2024). This can help to reduce financial costs and attract and retain customers while deepening community trust in the business (Mars, 2022; Radić et al., 2023). However, these relationships are more common with direct and local customers whereas relationships with larger scale customers tend to be more transactional and contractual than value-based, which can also lead to producers experiencing power imbalances and high dependencies (Belliggiano et al., 2024; Thomas-Francois et al., 2018).

In addition, suppliers are highlighted as important partners to ensure the supply of resources for production or harvest such as seeds or pesticides, but they also need to match producer requirements regarding organic, local or seasonal resources and production (Charatsari et al., 2023; Mars, 2015; Wiśniewska-Paluszak and Paluszak, 2021). Wiśniewska-Paluszak and Paluszak (2021) show that producers often work with local and personally well-known partners and long-term contractors, whereas a study by Mars (2015) highlights how producers also source from non-local suppliers with whom they not necessarily have a relationship with due to limited availability of local resources.

Further, cooperating with other producers, cooperatives, or branch associations, even if they are competitors, can positively affect value creation dynamics (Mars, 2015, 2022; Prosperi et al., 2023;

Wiśniewska-Paluszak and Paluszak, 2021). This can include pooling assets to reach joint payoffs, sharing of experience and knowledge of the territorial context, but requires to first establish a space of no-conflict (Prosperi et al., 2023; Wiśniewska-Paluszak and Paluszak, 2021). For example, a study by Lawson et al. (2008) shows how producers selling at a farmers' market engage in different forms of cooperation related to production, marketing or sharing equipment. However, studies also highlight the limited cooperation between producers when it comes to off-farm activities (Recasens et al., 2016), or when selling via wholesalers and food distributors due to cross-channel conflict, potentially limiting value creation opportunities (Lawson et al., 2008).

Finally, other actors such as neighbours, friends and the local community are highlighted as relevant for voluntary labour (Prosperi et al., 2023; Recasens et al., 2016), even though Recasens et al. (2016) found that there is otherwise little interaction between producers and these actors. Further, the cooperation with institutional actors such as advisory services, local governments, chambers of commerce, state agencies, universities and local action groups is recognised as important by scholars (Prosperi et al., 2023; Wiśniewska-Paluszak and Paluszak, 2021). For example, Prosperi et al. (2023) argue that policies can influence value delivery by for example encouraging quality certifications, and Recasens et al. (2016) highlight the role of public institutions to promote local products, facilitate access to land for new entrepreneurs, or simplify bureaucracy for subsidy applications.

3.1.3.5 Technologies used

Scholars mainly discuss the use of digital technologies in terms of better connecting producers and customers, and facilitating added value creation (Csordás et al., 2022; Musa et al., 2023). For example, digital platforms and social media can create value by increasing transparency, allowing interaction with numerous actors, and reaching a broader customer base without being limited to specific localities (De Bernardi and Tirabeni, 2018; Mars, 2022; Musa et al., 2023). In addition, Musa et al. (2023) highlight that digital platforms can allow producers to optimise key activities, maintain quality and better handle logistics, or act as intermediaries taking over some activities from individual producers, allowing them to focus on production rather than supporting activities. However, a study by Csordás et al. (2022) indicate that the majority of producers do not use any or very limited digital strategies to enhance their value creation and delivery.

3.1.4 Value capture

The value capture is influenced by the composition of the other BM components. For example, producers that base their value proposition on proximity to final customers can capture value by being able to set prices that reflect the true costs, or by having control over production (Albrecht and Smithers, 2018; Borsotto et al., 2023). Further, different distribution logics can have an impact on the costs and revenue structure (Belliggiano et al., 2024; Drejerska et al., 2019; Schutzbank and Riseman, 2013). Producer-led distribution may allow for direct payments and premium prices but can include market fees while buyer-led distribution can have the advantage of receiving upfront payment for the products but potentially to a discounted price (Borsotto et al., 2023; Drejerska et al., 2019; Schutzbank and Riseman, 2013). Further, buyer- or distributor-led distribution can require more support activities leading to higher costs while the income needs to be shared with other intermediary actors (Belliggiano et al., 2024; Drejerska et al., 2019). In addition, all indirect distribution channels require handling more stakeholder relationships affecting the value capture (Drejerska et al., 2019).

3.1.4.1 Cost structure

Costs discussed by scholars can be differentiated into three categories: First, cost related to the production at the local farms, including costs for material such as costs for seeds, plants or tools (Borsotto et al., 2023; Charatsari et al., 2023; Drejerska et al., 2019; Prosperi et al., 2023; Schutzbank and Riseman, 2013). Second, labour costs are often discussed by scholars, even though these typically only include non-family members. As labour costs are high, producers often do not account for their own labour in their expenses (Drejerska et al., 2019; Prosperi et al., 2023; Schutzbank and Riseman, 2013). Third, costs related to general business operations such as product transport to the points of sales, waste disposal, administration, marketing, and selling such as market fees (Borsotto et al., 2023; Charatsari et al., 2023; Drejerska et al., 2019; Schutzbank and Riseman, 2013). Scholars highlight that costs can fluctuate depending on external factors and can therefore have a great impact on the BM (Prosperi et al., 2023). Costs for land is also mentioned by some scholars (Pölling et al., 2017), while Schutzbank and Riseman (2013) argue that producers in the urban environment favour to provide other methods of payment for land access, such as trade for weekly produce or other non-monetary benefits.

3.1.4.2 Revenue streams

The revenue streams discussed can be classified into three categories. First, revenues related to selling food products and food production (Borsotto et al., 2023; Charatsari et al., 2023; Drejerska et al., 2019; Schutzbank and Riseman, 2013; Wubben et al., 2013). This can also include inputs from self-consumption or unpaid family-labour that support production and sales (Prosperi et al., 2023). Second, revenues related to services such as tourism and educational activities and third, financial support from public institutions through grants or, for example, by employing individuals who have difficulties on the regular job market or by engaging in care farming (Borsotto et al., 2023; Pölling et al., 2017; Prosperi et al., 2023; Rytkönen, 2016; Schutzbank and Riseman, 2013; Wiśniewska-Paluszak et al., 2023).

3.1.4.3 Value captured by key actors

Scholars discuss value captured by other food systems actors as well as regional economic impact of local food BMs (De Bernardi and Tirabeni, 2018; Borsotto et al., 2023; Moschitz and Oehen, 2020). For example, customers can capture value through consuming high-quality, authentic, transparent, healthy and safe products (Albrecht and Smithers, 2018; Drejerska et al., 2019; Keech et al., 2023). Drejerska et al. (2019) highlight that the value captured by customers can differ depending on the distribution logic of the local food BM, where producer-led distribution channels enable customers to capture value related to transparency and information about production processes, which may not be given in distributor-led channels, these however may allow for more convenience and easier accessibility. Further, engaging with various actors in the food system can also lead to various tangible and intangible benefits that can influence the value capture (Stoeva et al., 2024). For example, the local community and other companies located in the area can benefit from local food BMs as they potentially create new sources of income and employment,

new opportunities to diversify farming activities and increased resilience (Borsotto et al., 2023; Moschitz and Oehen, 2020).

Some scholars also discuss social, cultural and environmental value captured (Borsotto et al., 2023; Charatsari et al., 2023; Prosperi et al., 2023; Recasens et al., 2016). For example, a study by Charatsari et al. (2023) highlights the scale of outreach and the ability to diffuse value to local communities as most important for social impacts. Further, the design of distribution channels or activities can influence emissions and waste of a business and with that lead to greater environmental value being captured or more environmental value destroyed (Borsotto et al., 2023; Charatsari et al., 2023). In addition, the promotion of biodiversity through grazing or saving specific animal races, and the use of traditional agroecological methods can support the maintenance of agricultural and natural landscapes while generating higher revenues through for example visitors (Radić et al., 2023; Rytkönen, 2016).

3.2 Local food BM innovation

3.2.1 Preconditions for BM innovation

Scholars discuss several preconditions that influence a producer's innovative behaviour (Aggestam et al., 2017; Hersleth et al., 2023; Rytkönen and Oghazi, 2022). For example, studies indicate that the intention to scale up is driven by personal and social motivations, such as a positive attitude, personal priorities and a positive perception of family, customers or co-workers towards scaling the business (Aggestam et al., 2017; Rytkönen and Oghazi, 2022). Further, a producer's high intrinsic motivation for tradition, handcraft or sustainability has been highlighted as starting point for an innovative attitude, as well as entrepreneurial practices such as risk-taking and following the passion (Hersleth et al., 2023). However, a study by Aggestam et al. (2017) found that entrepreneurial skills, while being a necessity for producers to maintain a unique BM, are not the driving forces for scaling up the BM. In a similar vein, Alvarez et al. (2021) state that innovation strategies can be successful even with weaker entrepreneurial skills if the contextual factors, for example institutional protection provided by labels, are aligned with the managerial and technical skills of the producer facilitating innovation strategies.

Further, organisational and socio-economic characteristics such as a higher number of employees, limited dependency on infrastructure, size of farm, type of production, and when the business is the main source of income were found to foster the intention to scale up as well as the perception that it could reduce the potential of business failure (Aggestam et al., 2017; Fleiß and Aggestam, 2016). In contrast, lack of financial resources, need of investments, and for older producers, the unwillingness of the children to take over, were influencing the decision against innovations (Rytkönen and Oghazi, 2022). Also, producers who have worked in the field for a long time were found to have less intentions to scale the business (Fleiß and Aggestam, 2016). Perceptions of other producers and competitors as well as perception of external factors such as governmental support were neither found to influence the intention to scale the business (Aggestam et al., 2017).

3.2.2 Approaches for BM innovation

When aiming to innovate the BM, Rytkönen and Oghazi (2022) discuss that innovations are seldom disruptive, but producers tend to identify and exploit market opportunities based on imitating others, that is learning from and copying products and services. In line with this, scholars propose to experiment and pilot with various strategies to find the best approach and ensure viable BM innovation (Errickson et al., 2022; Vidickiene et al., 2023). For example, Vidickiene et al. (2023) emphasise BM design thinking to explore and test BM innovation. This includes to learn about and better understand customers values and how they align with producers' needs, define challenges of the current BM, create ideas how to overcome the identified challenges, and then test various prototype innovations. This is of importance as any BM innovation typically requires high investment and comes with high risk (Royer and Simon, 2023).

3.2.3 Strategies for BM innovation

A variety of different strategies to innovate the value proposition and value creation and delivery are discussed by scholars.

3.2.3.1 Diversify and customise value proposition elements

Scholars argue that producers need to better differentiate the value proposition by offering various products with distinctly different product characteristics such as local, quality, organic, good value for the price, or by highlighting specific benefits of the products such as seasonal harvest (Recasens et al., 2016; Rytkönen and Oghazi, 2022). In line with this, Hersleth et al. (2023) argue that producers need to use differentiation strategies to legitimise the superior quality of their product and follow a personal market approach that make the innovation hard to copy. This can be achieved by being proactive, pushing for new approaches through networking, building strong customer relationships, by getting endorsement from famous chefs or food competitions, or by using labels (Hersleth et al., 2023). However, while this strategy can enable to clearly differentiate the value proposition from other products, sell outside their region, get higher prices and a less dependent position in the market, it is also a time-consuming process and can require a more resource intense production (Hersleth et al., 2023; Radić et al., 2023; Royer and Simon, 2023).

Further, customising the value proposition to specific customer segments is highlighted by scholars as an essential process (Wiśniewska-Paluszak et al., 2023). This does not only include to develop value propositions that cater customer interest but value propositions where customers understand the overarching value of local food products and services (Sebök et al., 2022). This can be achieved by better targeting specific issues such as food safety and quality, trust, sustainability, ethical aspects or accessibility to enhance its core value and improve competitiveness (Sebök et al., 2022). These values need to be clearly communicated, otherwise it may lead to "wastage of value" where customer cannot differentiate between conventional and local food if an intermediary organisation is involved (Thomas-Francois et al., 2018, p. 412). The match between the value proposition and customers is thus the base to be able to grow and develop the business (Wiśniewska-Paluszak et al., 2023).

3.2.3.2 Upgrade, diversify and integrate new activities

Several scholars discuss value creation and delivery innovation strategies regarding how producers can add value to their primary production activities (Alvarez et al., 2021; Royer and Simon, 2023). That is because producers often utilise the already existing farm-based

resources in additional business activities in innovative ways (Alvarez et al., 2021; Hersleth et al., 2023). For example, scholars discuss to innovate processes and packaging activities by intensifying production of a single product, getting a specific production label, or increasing the number of products produced which can lead to improved efficiency and reduction of production costs (Alvarez et al., 2021; Royer and Simon, 2023). Further, producers can increase the product range or product sizes, start with personalised production or innovative and memorable packaging and refill options to enhance value creation (Alvarez et al., 2021; Charatsari et al., 2023; Hersleth et al., 2023; Radić et al., 2023). Alvarez et al. (2021) argue that these strategies do not require high levels of entrepreneurial skills or proactiveness as producers can focus on their original products and established markets, but it may be difficult for producers to differentiate themselves from other producers due to the minimal transformation of their production.

Producers can also diversify their activities by combining traditional production activities with innovative and non-agricultural activities such as engaging in tourism or education, including visits from schools to teach biological or technical processes or guided tours (Alvarez et al., 2021; Pölling et al., 2017; Recasens et al., 2016; Rytkönen and Oghazi, 2022; Vidickiene et al., 2023; Wiśniewska-Paluszak and Paluszak, 2021). Further, leisure and cultural activities aimed at spreading knowledge, tasting and culinary activities, or activities related to the landscapes and buildings are discussed (Pölling et al., 2017; Recasens et al., 2016). The integration of such activities can allow to create an emotional link between product and customers while promoting farm activities (Di Gregorio, 2017; Rytkönen and Oghazi, 2022). In particular, tourism BM diversification can be linked to various production or processing activities and can be implemented on various levels, from focusing on tourism as a secondary activity to making tourism the primary activity while farm activities become secondary (Di Gregorio, 2017; Recasens et al., 2016). These activities can in addition to increased economic value capture also strengthen the social function of local food production (Recasens et al., 2016; Vidickiene et al., 2023). However, the implementation of such activities may also create more complex customer needs that producers must address and, when collaborating with tourism actors, may require infrastructure development (Belliggiano et al., 2024; Vidickiene et al., 2023).

In addition, some scholars discuss integrating new activities from other actors in the supply chain such as starting with product elaboration or selling and marketing products in new ways (Royer and Simon, 2023; Rytkönen and Oghazi, 2022). For example, study by Rytkönen and Oghazi (2022) focused on the dairy industry highlights a shift from the traditional separation of farming activities, where the elaboration of farm products was conducted by large companies, towards producer-owned, small-scale cooperative dairies that enabled producers to elaborate standardised products. Royer and Simon (2023) refer in this context also to intersectoral innovation where producers integrate other supply chains, for example, by including other animals or plants on their farms, establishing own energy supply or starting agroforestry projects. Taking on such new activities can lead to a more powerful position in the supply chain (Royer and Simon, 2023).

3.2.3.3 Expand resources

Some scholars discuss strategies to expand resource for value creation. For example, one strategy proposed is to include more volunteers as a labour resource during harvest as well as in other activities such as tourism or recreational activities (Recasens et al., 2016). Further, Recasens et al. (2016) discuss changing from monoculture to polyculture via crop diversification enabling producers to produce a wider range of products. However, the scholars also highlight that only a few producers in their study produce other agricultural products than their main product (Recasens et al., 2016). Further, acquiring and diversifying knowledge such as developing digital skills are highlighted as important (Oleksiuk and Rull Quesada, 2023). This enables producers to be involved in all production stages, needed to innovate the BM (Radić et al., 2023).

3.2.3.4 Diversify distribution channels

Another strategy explored by some scholars is to add new and diversify distribution channels, for example to sell products to local schools, hospitals and municipal institutions, or via vending machines located in various local institutions or in customer proximity (Oleksiuk and Rull Quesada, 2023; Pereira et al., 2018). However, scholars also highlight various barriers and complexities with establishing such a new distribution channel, for example the need for infrastructure development, more marketing or public procurement and the potential lack of product variety and lack of product availability in the new sales channel (Oleksiuk and Rull Quesada, 2023; Pereira et al., 2018). Further, Vidickiene et al. (2023) test service-oriented distribution channels such as pre-ordering systems with free choice of products or made-to-order systems with subscription of standardised products, arguing that the first can be very labour intensive and difficult to manage while the latter may not be sufficiently innovative.

3.2.3.5 Collaborate with specific actors and via intermediaries

Several scholars discuss to innovate the BM by collaborating with specific key actors. One approach is to collaborate with other producers to innovate specific activities or distribution channels by sharing logistics to the points of sale, jointly opening a storefront to sell products, or jointly leasing elaboration and packaging from larger producers to reduce costs for investments and employer liability (Oleksiuk and Rull Quesada, 2023; Rytkönen and Oghazi, 2022). For example, Mars (2015, 2022) show how a local producer through partnerships with other producers and local food actors developed a new local food supply chain. Further, Errickson et al. (2022) explore how producers and community organisations work together to develop a collaborative BM, creating new opportunities to capture economic value while reducing food waste and offering educational training. Partnerships with other actors in the food system such as specialty shops, local hotels, restaurants and chefs, or fairs, events, social movements as well as cooperation with tourism institutions can help producers to better engage with customers, establish new distribution channels, enhance visibility of products and facilitate access to new markets (Radić et al., 2023; Stoeva et al., 2024). Scholars also argue for engaging with knowledge sources outside the local community for innovation processes, such as research institutions, agricultural experts or general advisors to learn how to develop and strengthen local food BMs (Moschitz and Oehen, 2020; Rytkönen, 2016).

However, effective cooperation and creation of joint opportunities require a lot of coordination, adaption of farming, processing and

market positioning, as well as building of trust and relationships, in particular with actors that have previously not been involved (Belliggiano et al., 2024; Mars, 2022; Oleksiuk and Rull Quesada, 2023). Errickson et al. (2022) further discuss challenges of matching needs and production capacities of all actors involved while yielding economic benefits. One negative example is provided by Pereira et al. (2018) who analyse a collaborative start-up of several producers that failed due to high investment costs, lack of competence, political support, long-term shared vision and lack of collaboration with other stakeholders such as intermediaries limiting the potential to scale-up product diversity.

To collaborate via an intermediary is discussed by several scholars, for example by establishing or joining an existing cooperative (Hooks et al., 2017; Moschitz and Oehen, 2020; Rytkönen, 2016). Rytkönen (2016) differentiates between structural cooperative approaches, encompassing the foundation of a cooperation that facilitates joint distribution and common warehouses, and functional approaches focused on identification and qualification of territorial assets facilitating joint production innovation. This can positively influence the value creation dynamics of individual local food BMs, opening new sources of income, new ways to cooperate with other businesses, and new opportunities to diversify activities and distribution channels, reaching a broader range of customers (Di Gregorio, 2017; Moschitz and Oehen, 2020; Rytkönen and Oghazi, 2022). However, Hooks et al. (2017) also highlight the risk of cooperatives having too broad geographical expansion, potentially requiring more certifications and leading to spatially diffused members and reduced sense of identification and commitment.

Further, scholars discuss collaboration via an intermediary outside the direct food chain, for example, Braun et al. (2023) explore how a learning platform can create formal patterns of collaboration for producers and other actors in the agri-food sector to share knowledge and facilitate new value creation strategies. In a similar vein, Charlebois and Mackay (2010) discuss the possibility of creating a regional-based organisation that explicitly ties products with local culture, helping producers to develop new business opportunities, add value to existing products, or develop a portfolio of different value delivery channels. However, these kinds of approaches need to be actively developed and embedded in a local community to address specific challenges (Braun et al., 2023; Charlebois and Mackay, 2010).

3.2.3.6 Add new technologies

Scholars highlight the large potential of innovation strategies with focus on digital technologies such as digital marketing or platforms, providing opportunities to shorten the food supply chain, scaling up the local food BM and increasing the resilience of the business (Csordás et al., 2022; Musa et al., 2023). Some scholars also discuss the use of blockchain technology in the supply chain as a governing solution to create new local food BMs (Oleksiuk and Rull Quesada, 2023). Based on a collaborative approach, blockchain technology could enhance the traceability and transparency of production for customers, facilitating better flow of information, collaboration and coordination between producers and more cost-efficient distribution of product delivery, allowing all actors to become an integral of the local food system (Oleksiuk and Rull Quesada, 2023; Sekuloska and Erceg, 2022). However, scholars highlight various challenges with the adoption of digital technologies, among others the lack of technological expertise, financial resources and willingness of stakeholders to support such projects as well as privacy issues (Musa et al., 2023; Sekuloska and Erceg, 2022).

4 Concluding discussion

4.1 Main findings and implications

Local food BMs have been recognised as a crucial means to diversify markets and redesign food systems. Yet most research has treated them as a sub-category of sustainable food BMs or explored them in isolated studies that emphasise the uniqueness of individual cases (Prosperi et al., 2023; Rytkönen and Oghazi, 2022). This review takes a first step towards organising the fragmented literature, establishing an integrative perspective by proposing the local food BM and innovation framework (Figure 3).

Thus, this review contributes to three main insights. First, it systematises the conceptual and empirical literature on local food BM components, offering in-depth understanding of what the local food value intention, proposition, creation and delivery, and capture encompass. While value intentions are often economically driven, they frequently integrate social and environmental dimensions, shaping how value propositions are designed and combined. Four distinct value proposition elements were identified, proximity and place-making values, product characteristics values, intangible values, and sustainability values, closely aligned with common definitions of local food production such as social and geographical proximity and sustainability (Eriksen, 2013; Nilsson, 2009; Pardillo Baez et al., 2020). The review further highlights various configurations and the interdependence of key activities, resources, distribution channels, key partners and relationships, and technologies used in value creation and delivery, as well as their advantages and trade-offs. Importantly, the effectiveness of value propositions is contingent on these dynamics (Prosperi et al., 2023; Wubben et al., 2013), which also determine how and what kind of value is ultimately captured (Drejerska et al., 2019; Wiśniewska-Paluszak et al., 2023).

Second, the review demonstrates how local food BMs can be innovated, highlighting preconditions for BM innovation, such as producers' motivation to scale and willingness to embed sustainability in their value proposition (Barth et al., 2017). Innovation approaches are often iterative and imitative, ensuring feasibility and long-term sustainability. Ten strategies were identified for innovating the value proposition and value creation and delivery, with strong relationships and collaborative action emerging as a key factor (Fobbe and Hilletofth, 2021). These strategies, while presented separately are rarely implemented in insolation; rather, they are interconnected, for example, diversification of activities can facilitate the integration of new activities. However, scaling and innovation also demand additional resources, including new skills, time, and investments, highlighting the complexity of BM innovation and its systemic effects across all BM components (Clauss, 2017; Royer and Simon, 2023; Rytkönen, 2016).

Third, the framework developed in this study underscores that successful BM innovation requires producers to fundamentally understand the type of value they provide, the mechanism of value creation and delivery, and the range of options for meeting customer needs. This review thus offers a first steppingstone for both scholars and practitioners to systematically analyse, develop, and innovate local

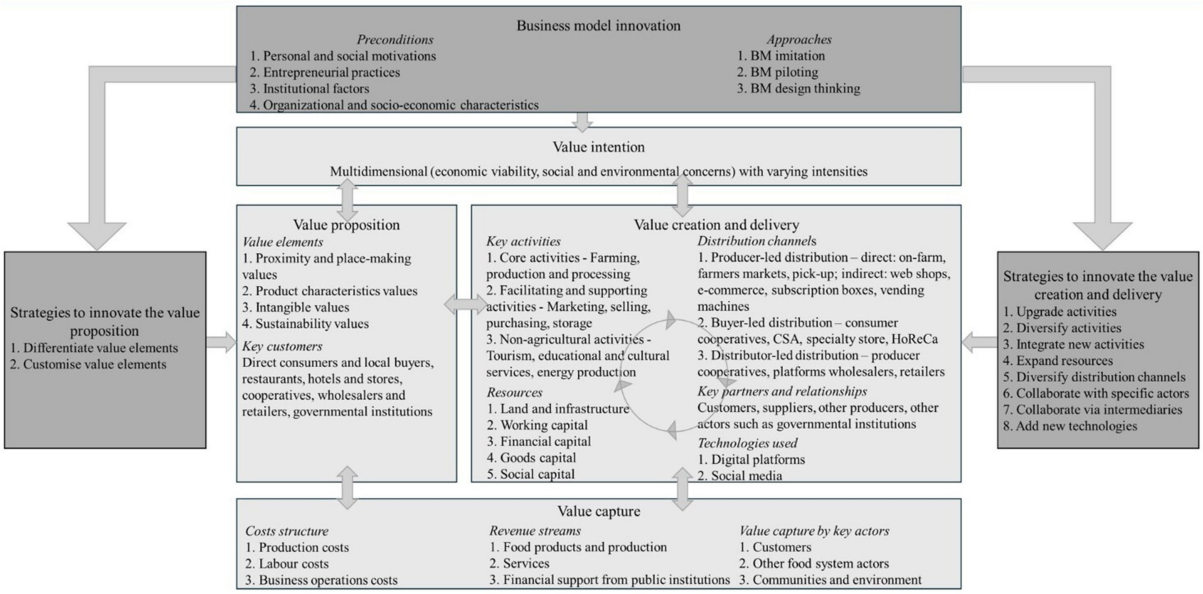


FIGURE 3
Local food business model and innovation framework.

food BMs. In sum, this study provides a comprehensive understanding of local food BM components and innovation strategies, offering new insights into how local food production can be fostered and scaled. It contributes to sharpening scholarly debates on the role of BM perspectives in sustainable food system research and advances discussions on BM innovation and local food businesses. The proposed local food BM and innovation framework can provide practitioners with knowledge and actionable guidance on how to understand and approach local food BMs and BM innovation, revealing different strategies to strengthen local food businesses. This can help to overcome institutional barriers, expand markets, and ensure long-term viability. For policymakers, the findings highlight new opportunities to design supportive frameworks and incentives that strengthen local food production. Taking this together, these insights underline the importance of BM thinking in supporting the transition towards more sustainable and resilient food systems.

4.2 Limitations and future research directions

The framework developed in this paper highlights the multidimensional and reciprocal relationships between local food BM components and BM innovation strategies, offering new insights into value generation mechanisms. Nevertheless, several limitations should be noted. First, this study focused on local food BMs from a producer's perspective, but excluded studies explicitly focused on other types of local food BMs such as intermediaries which can play an important role in local food systems. Second, this review provides a comprehensive perspective of the logic of how local food BMs operate and how they can be innovated, but several articles may have been published related to the topic after conducting it. Therefore, some relevant publications on this topic may not be included in this review. Third, this study only included peer-reviewed articles to ensure high-quality research but excluded conferences papers and books as well as articles in languages other than English might have provided further valuable insights.

Based on the findings of this review, five research directions for further investigation have been identified. First, linked to the understanding of local food BMs, this review identified what the value components of local food BMs encompass and highlighted their interlinkages and trade-offs. But this review also showed that some specific elements in local food BMs warrant further investigation. For example, this review highlights the importance of digital technologies for local food BMs and BM innovation, but it also indicates that few producers currently exploit its potential. This is a crucial gap to bridge; thus, a first potential research avenue is to explore the digital maturity of local food producers and strategies to advance their digitalization. Second, strong relationships and collaboration with various actors were found to be a key element of all BM components and BM innovation strategies. However, most research in this review is focused on the role of customers and their engagement, while there is less on other actors such as other producers, intermediaries, or other actors and their role for local food BM innovation. Future research could explore how such collaborations can be established and organised, and how they can contribute to fostering local food BMs and, in the long-term, local food systems. Third, most studies in this review had a qualitative character, while there is a lack of quantitative studies. Future quantitative research could test the relations and interlinkages within and across local BM components and their implications.

Further, linked to local food BM innovation, this review identified preconditions, approaches and 10 strategies to innovate local food BMs. While interlinkages between the strategies are highlighted, there is still a lack of knowledge how producers can combine the different strategies. Thus, a fourth research direction could be to investigate

how the choice of strategies may depend on preconditions and intentions to scale, and which factors can influence the success of implementing different scaling strategies. Fifth, most studies included in this review focus on incremental BM transformation of individual BM components with only a few studies discussing starting-up new or diversifying BMs (Geissdoerfer et al., 2018). This may explain why there is limited growth of local food production and indicates a potential need for further research, exploring how to approach more radical transformations of local food BMs to speed up moving towards sustainable and resilient food systems.

Data availability statement

The original contributions presented in the study are included in the article/Supplementary material, further inquiries can be directed to the corresponding author.

Author contributions

LF: Writing – original draft, Writing – review & editing. PH: Writing – review & editing.

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Conflict of interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Supplementary material

The Supplementary material for this article can be found online at: https://www.frontiersin.org/articles/10.3389/fsufs.2025.1703973/full#supplementary-material

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